



Combine these three documents in a structure that conveys the following concepts: 1. executive summary explaining how this platform is a simple NL window to provide a panoramic view of many datasources. 2. How to use this platform for teaching AI engineering (and icing hundreds of hours of prompt engineering, coding standards, and architectural discipline supported by a personal tutor: Gemini 2.5 pro) 3: featuring enhancements are possible by scaling up the current data to all ports of main besides Machias; other subject matters like ecological analysis of seawater sensors; most importantly, adding images additional data format

The Astoria Open platform offers a unified conversational AI interface that presents a panoramic view of structured and unstructured data from diverse sources using natural language. Designed to facilitate both historical inquiry and AI education, it integrates advanced architectural and pedagogical innovations, while being open for future scaling and multimedia enhancements.^[1]
^[2] ^[3]

Executive Summary: A Panoramic NL Window Across Data Sources

Astoria Open delivers a **simple, natural language (NL) portal** through which end-users can query and explore disparate information repositories—ranging from traditional SQL databases to unstructured document libraries—with remarkable fluency. The system’s React-based interface allows users to pose questions in everyday English. Astoria’s backend, powered by FastAPI on Gunicorn, orchestrates complex data retrieval and synthesis: direct factual queries are instantly translated into SQL, while the system simultaneously semantically searches a vectorized document archive. The resulting narrative fuses these varied strands, offering **transparent, verifiable, and easily navigable insight** into the entire knowledge domain—for example, Maine maritime history. The underlying open-source architecture is designed for **efficient, concurrent multi-user access** and minimizes duplication by leveraging shared backend resources for all users.^[2] ^[3] ^[1]

AI Engineering Pedagogy: Transparent, Guided Learning Environment

Astoria Open is more than a tool for inquiry—it is a platform for teaching **current AI engineering practices**. The architecture, source code, and documentation are fully accessible. Trainees interact with the running application, reverse engineer its components, and learn through hands-on experimentation supported by the Gemini 2.5 Pro model as a **domain-personalized AI tutor**. This tutor can explain application dataflows, architectural decisions, and prompt engineering strategies, providing a view into hundreds of hours of best practices in:^[3] ^[2]

- **Prompt Engineering:** See how natural language gets transformed into actionable SQL or retrieval tasks.
- **Coding Standards:** Examine code designed for maintainability, transparency, and scalability.
- **Architectural Discipline:** Observe modern patterns such as stateless load balancing, in-memory shared models, and separation of concerns in production architecture. Learners can inspect how their queries are routed, how vector embeddings work, and even review the raw SQL and AI-generated responses. By making every part of the pipeline visible, the experience is akin to “watching a transparent engine run,” thus demystifying AI systems for educators, students, and community members alike.^[2] ^[3]

Feature Expansion: Data Scaling, Subject Flexibility, and Multimedia

The platform’s flexible, modular design enables straightforward scaling and domain adaptation:

- **Expanding Historical Context:** Beyond the current comprehensive Machias maritime records, Astoria’s infrastructure supports the easy inclusion of million-scale datasets—such as logs from all Maine ports, state-wide historical documents, and archival images.^[3] ^[2]
- **New Domains:** The Hub-Spoke architecture generalizes to domains like ecological sensor networks, where structured sensor data and scientific reports can be queried and synthesized just as seamlessly as maritime data.^[2] ^[3]
- **Enhanced Data Formats:** The next phase includes integrating **images and other multimedia formats**, leveraging open-source multimodal LLMs and image libraries. This will allow users to query visual resources, view maps, schematics, or artifacts, and receive narrative summaries or analytic insights that feature both text and images—significantly enriching results and user experience.^[3] ^[2]

The pathway forward is both practical and vision-driven: by expanding the scope of supported datasets and modalities, while maintaining the core ethos of open-source transparency and educational accessibility, Astoria Open aims to continue democratizing AI and making rich historical, scientific, and educational resources available to all.^[1] ^[2] ^[3]



1. Astoria-Open-multisuer.pdf

2. Astoria-Open-Architecture-v2.pdf

